MANIPAL INSTITUTE OF TECHNOLOGY

Manipal Academy of Higher Education, Manipal

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

THIRD SEMESTER B.TECH. END SEMESTER EXAMINATION - 2025

CS 2105: DISCRETE MATHEMATICAL STRUCTURES

Branch: Computer Science & Engineering (CSE) | Credit: 4

Time: 3 Hours | Max. Marks: 50

Instructions: Answer ALL questions. Each question carries designated marks.

PART A

1. (a) State and prove Principle of Mathematical Induction for sum of squares of first n integers. [5 Marks]

1. (b) Differentiate equivalence relations, partial orders, Posets, and Hasse diagrams. [5 Marks]

PART B

2. (a) Solve linear recurrence relation: a_n = 5a_{n-1} - 6a_{n-2} with a_0 = 1, a_1 = 4. [5 Marks]

2. (b) Explain Pigeonhole Principle with engineering examples on hash bucket collisions. [5 Marks]

PART C

3. (a) State Euler theorem for planar graphs and prove K_5 and K_3,3 are non-planar graphs. [10 Marks]

PART D

4. (a) Explain Group theory: Monoids, Semigroups, Abelian Groups, and Lagrange Theorem on subgroups. [10 Marks]

[END OF QUESTION PAPER - MIT MANIPAL CSE 3RD SEMESTER ARCHIVE]